



Intercept Carbon

Carbon dioxide removal prepurchase application
2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Intercept Carbon

Company or organization location (we welcome applicants from anywhere in the world)

Seattle WA

Name(s) of primary point(s) of contact for this application

Christoph Mack

Brief company or organization description <20 words

ThermoSonic Carbonization of wet organic wastes & sequestration of solid carbon in mine voids at low cost and energy consumption.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

1. Introduction

Intercept Carbon's technology converts wet organic wastes into Hydrothermally Activated Carbon (HAC) pellets and nutrient-rich liquid fertilizer. Our first focus is on the conversion of dairy manure – one of the most methane-intensive organic waste streams on earth. The resulting dry, carbon-dense, highly aromatized solids (Hydrothermally Activated Carbon, or "HAC") are then placed into abandoned mine voids for permanent entombment.

The system is deployed through regional processing hubs. One 80 tonne/day ThermoSonic Carbonization (TSC) facility serves ~8 medium-sized dairies, leveraging the natural clustering of dairies in the U.S. (85% within 5 miles of another, 95% within 10 miles) to minimize logistics costs and transport emissions.

Our approach delivers:

- Direct carbon removal via conversion of biogenic carbon into stable aromatic solids.
- Durable storage through both chemical stability and geologic entombment.
- Methane avoidance as a co-benefit (critical for climate impact, though we recognize Frontier evaluates only direct removals in its formal accounting).
- Superior farmer economics, paying up to \$40 per dry tonne while returning concentrated nutrients.

2. How CO₂ is Removed from the Atmosphere

2.1 Capturing Biogenic Carbon

Cow manure embodies carbon fixed by plants through photosynthesis. In status quo management (lagoons, untreated land application), this carbon re-enters the atmosphere rapidly as CO₂, CH₄, and N₂O. By intercepting manure at the farm gate, we capture and stabilize this carbon before it re-emits.

2.2 Conversion to Stable HAC

In the TSC reactor, manure slurry is processed at ~260 °C and ~40 bar. Within minutes, it undergoes dehydration, decarboxylation, and polymerization reactions. Traditional hydrothermal processes require long reactor dwell times (i.e. 4-12 hours) leading to large and expensive processing facilities.

The outputs of our TSC system:

- HAC solids (~20% of wet input).
- Nutrient-rich process water, later concentrated and returned to farms.

The HAC is then chemically and thermally activated into HAC pellets with:

- 70% fixed carbon.
- Low H/C and O/C ratios (high aromaticity).
- Moisture <22% (pellet-stable for storage).

The material is analogous to anthracite or activated carbon: chemically inert, microbially resistant, and built to persist for millennia.

2.3 Storage in Mine Voids

Pellets are delivered in roll-off boxes and poured into sealed abandoned mines. Unlike CO₂ injection, no pressure is required. The storage mechanism is:

1. Intrinsic permanence of aromatized HAC (>1,000-year stability).
2. Geologic entombment inside mine voids already sealed and monitored under closure regulations.

This dual stability mechanism ensures durable storage beyond Frontier's >1,000-year threshold.

3. Why Our System is Best-in-Class

3.1 Direct Removals First, Methane Avoidance as Co-Benefit

We explicitly recognize Frontier's framework: only direct removals count toward CDR accounting. Our direct pathway is clear: plant carbon → manure → HAC pellet → mine storage.

However, our approach also yields a significant methane avoidance co-benefit, because manure is diverted from lagoons and untreated land application. This does not alter Frontier's ledger but does substantially increase real-world climate impact per tonne removed.

3.2 Permanence and Safety

- Biochar competitors: O/C ratios often >0.5, leaving vulnerability to weathering and microbial turnover. High energy requirements.
- Bio-oil injection (Charm Industrial): Stores carbon in salt caverns, durable but reliant on geologic formations and logistics of liquid handling. Requires low-moisture feedstocks with competing alternate uses.
- DAC/mineralization: Durable but costly (\$600-1,000/tCO₂e today) and energy intensive.

By comparison, HAC pellets in mines:

- Offer permanence similar to geologic CO₂ storage.
- Pose very low risk of leakage, migration, or induced seismicity.
- Are simple solids at ambient conditions, requiring only trucks and roll-off boxes.

3.3 Farmer Economics

- We pay up to \$40/dry tonne of manure.
- Nutrients are extracted, concentrated, and returned as precision fertilizer, ensuring farms retain control of soil fertility.
- For small-to-medium dairies, participation can more than double net profits, far outpacing any economic benefit from AD or composting.

This alignment of incentives creates durable partnerships with agriculture, a scaling advantage no other CDR pathway can claim.

3.4 Simplicity and Scalability

- No pipelines, compressors, or Class VI permits.
- Leverages existing mine voids and dairy infrastructure.
- Short-haul logistics enabled by natural dairy clustering.

This makes ours one of the most practically deployable gigaton-scale CDR approaches.

4. Differentiation vs. Other Organizations

- Charm Industrial: Injects bio-oil into salt caverns. Strong permanence but constrained to suitable caverns and liquid logistics. Requires special handling, pumping, and regulatory clearance for cavern injection. Requires low moisture feedstocks for torrefaction.
- Graphyte: Bale-based biomass burial. Less chemically durable; significant risk of degradation and reversal.

- Carba: Soil biochar. Offers co-benefits but lower confidence of permanence.
- DAC companies (Climeworks, Heirloom, etc.): Extremely durable storage but cost and energy requirements far higher.

Intercept's unique combination:

- Direct removals with >1,000-year stability.
- Methane avoidance as a material climate co-benefit.
- Farmer-aligned economics that transform waste management into profit.
- Safe, solid-phase geologic storage requiring no new high-risk infrastructure.
- Low cost of processing, minimal energy and water consumption.

5. Alignment with Frontier Innovation Priorities

1. Biomass Carbon Removal with Storage (BiCRS):
 - Converts unavoidable wet manure into a durable carbon solid.
 - Avoids counterfactual emissions while achieving permanent storage.
 - Additional and scalable.
2. Durable Storage Innovation:
 - Introduces solid-phase pellet geologic storage, a novel variant to complement liquid bio-oil or supercritical CO₂ injection.
 - Leverages existing mine infrastructure with lower complexity, cost, and risk.

6. System Schematics



Figure 1. Intercept Carbon FOAK Hub Processing Facility Model

1. Dairy farm → roll-off box collection.
2. Short-haul transport (5 miles each way).
3. TSC + activation → HAC pellets.
4. Nutrient extraction → liquid fertilizer returned to farms.
5. Pellets trucked to sealed mine voids.
6. Placement + regulatory monitoring.

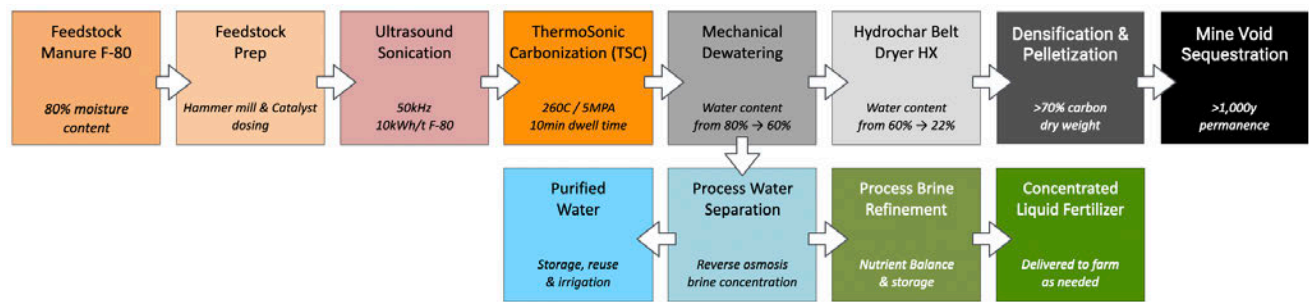


Figure 2. Intercept Carbon FOAK Process Block Diagram

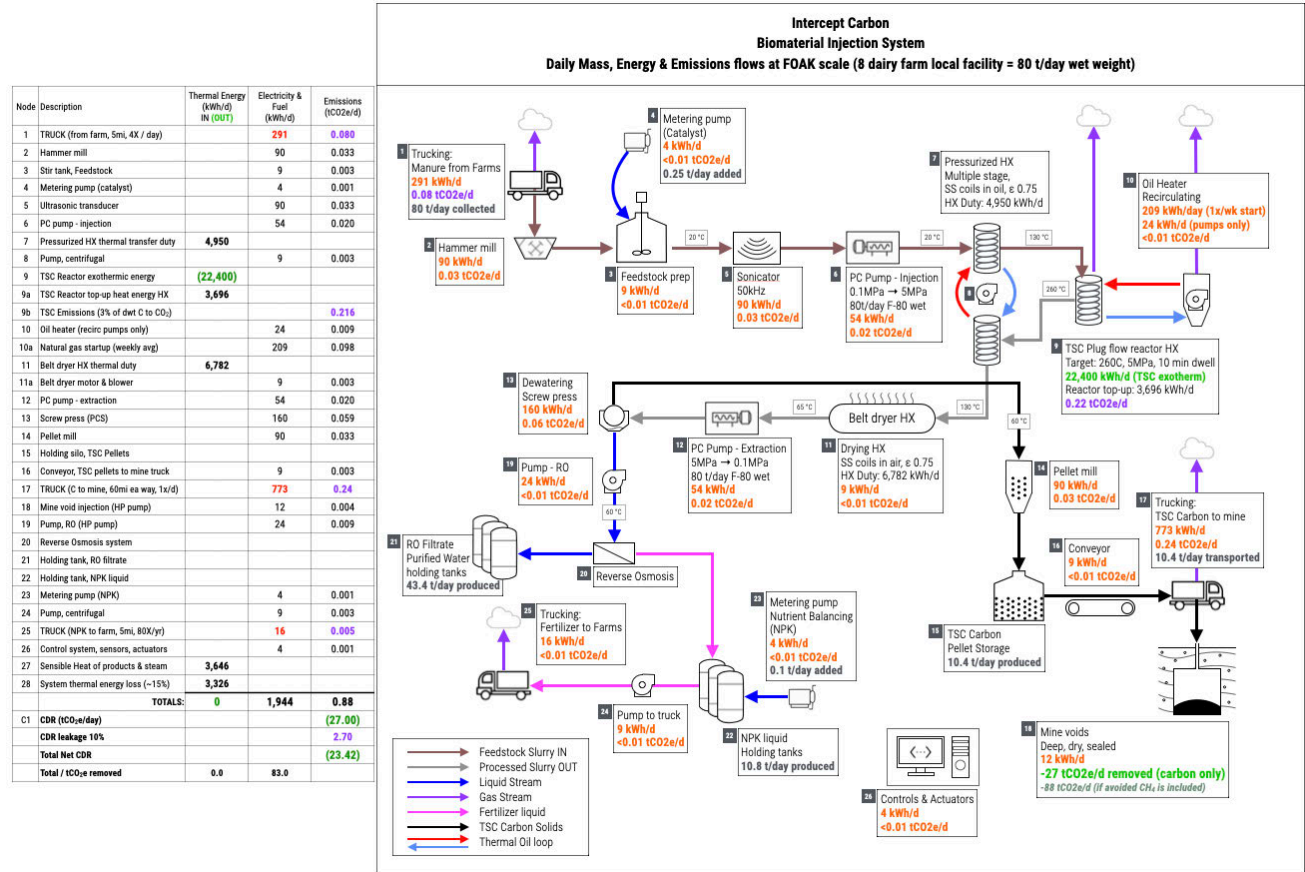


Figure 3. Intercept Carbon FOAK Process Schematic with Energy, Mass, and Emissions flows

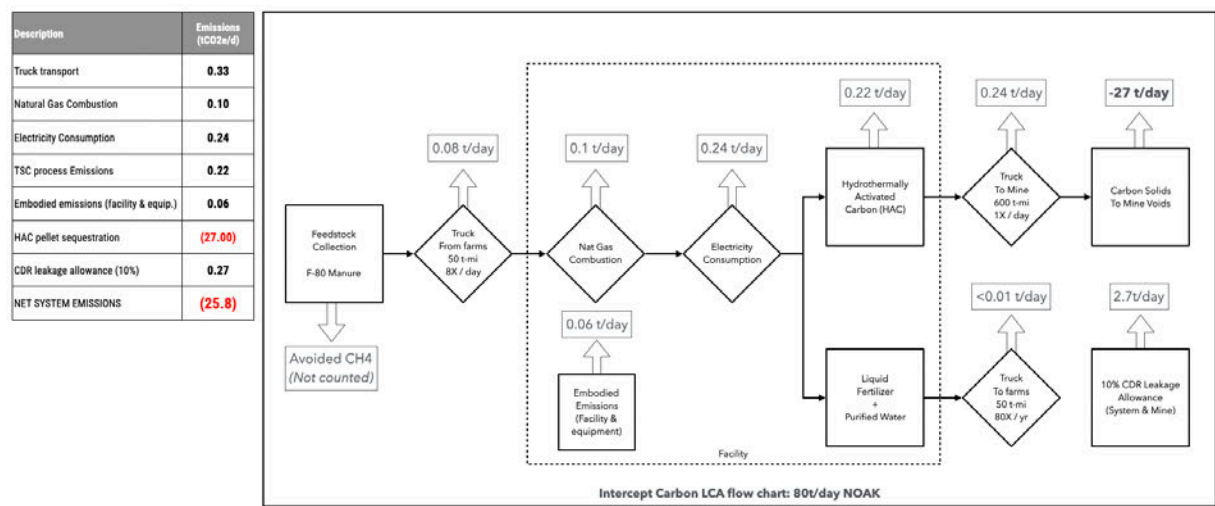


Figure 4. Intercept Carbon FOAK Process LCA Schematic (including 10% CDR leakage allowance)

7. Quantitative Performance

- Feedstock: 80 tonnes/day wet manure (~8 dairies).
- Output: ~10.4 tonnes/day HAC pellets (~70% fixed carbon).
- Direct CDR: ~27 tCO₂e/day → ~10,000 tCO₂e/year.
- Methane avoidance (co-benefit): Significant but tracked separately from Frontier’s accounting.
- Transport emissions: 0.5-0.8 kg CO₂/tonne manure (<0.1% of removals).
- Cost target: <\$60/tCO₂e at NOAK scale.

8. Monitoring and Verification

- Mass balance: Precise weighing of feedstock, pellets, and stored volumes.
- Geochemical tests: Proximate/ultimate analysis confirming aromatic stability.
- Mine monitoring wells: Routine groundwater quality checks (pH, conductivity, trace metals).
- Third-party audits: Verification consistent with mine closure frameworks.

This ensures independent verification of durability, placement, and absence of leakage.

9. Summary

Intercept Carbon’s approach – dairy manure → HAC pellets → mine entombment – delivers best-in-class permanence, scalability, cost, and farmer alignment.

We recognize that Frontier counts only direct removals; our accounting reflects only the stable HAC pellet carbon stored underground. Still, we note methane avoidance as a major real-world co-benefit, amplifying impact even if not credited.

Against biochar, bio-oil injection, and DAC, we combine permanence with simplicity, economic alignment, and rural development. By leveraging dairy clustering and underutilized mine voids, we unlock gigaton-scale potential with low cost and low complexity – a pathway that is additional, durable, verifiable, and widely replicable.

- b. Risks: What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Every carbon removal approach carries risks. We are transparent about the uncertainties we face, and we design our system to mitigate them. Below we describe risks across technical, execution, MRV, ecosystem/social, and financial domains, along with the mitigation strategies we have in place.

1. Technical Risks

1.1 Reactor Performance and Reliability at larger scale

Risk: Our Thermo-Sonic Carbonization (TSC) reactors must reliably process high-moisture wet organic feedstocks (such as manure) into stable HAC. Variability in manure composition (e.g., solids content, bedding material) could affect yields or system stability.

Mitigation:

- Extensive bench-scale testing has confirmed TSC works across a range of manure feedstocks.
- Pilot-scale demonstrations will validate throughput under real-world conditions.
- Automated feedstock conditioning (grinding, dilution) ensures consistent slurry properties.
- Parallel reactor modules allow for redundancy; if one unit requires maintenance, others continue operation.

1.2 Pellet Quality and Stability

Risk: HAC pellets must remain highly aromatized and structurally stable at ~22% moisture. Inadequate activation could reduce permanence.

Mitigation:

- Proximate and ultimate analysis of every production batch confirms H/C and O/C ratios (<0.5).
- Pellets undergo accelerated aging and leachate tests to validate stability.
- Pelletization process is standardized, ensuring uniform size and density.

1.3 Mine Storage Integrity

Risk: Decommissioned and abandoned mines voids vary in stability. Concerns include void collapse, groundwater ingress, or inadequate sealing.

Mitigation:

- Partner with mine engineers (e.g., SCS, HydroGeoLogic) to survey and qualify sites.
- Select only sealed, stable voids with existing monitoring infrastructure.
- Placement requires no pressurization; solids simply rest in voids, reducing risk of migration.

2. Project Execution Risks

2.1 Logistics and Collection

Risk: Inefficient collection and hauling could raise costs, emissions, or farmer dissatisfaction.

Mitigation:

- Use roll-off box swaps, minimizing on-farm burden.
- Leverage dairy clustering: 85% of dairies within 5 miles of another, 95% within 10 miles.
- Optimize routing with logistics software; begin at hub scale (8 farms) to validate.

2.2 Farmer Participation

Risk: Farmers may hesitate to commit to long-term manure offtake contracts.

Mitigation:

- Offer payments up to \$40/dry tonne, more than doubling profits for many dairies.
- Guarantee return of nutrients as liquid fertilizer, ensuring no agronomic loss.

- Structure agreements as co-op models to build collective trust and buy-in.

2.3 Mine Access and Permitting

Risk: Storage requires regulatory approval under Class V experimental permits. Timelines may vary.

Mitigation:

- Engage regulators early to establish precedent for solid carbon placement.
- Begin with demonstration in cooperative mine sites where environmental oversight is already robust.
- Develop standardized siting and permitting templates for replication.

3. Measurement, Reporting, and Verification (MRV) Risks

3.1 Carbon Accounting Accuracy

Risk: Over- or under-estimating net removals would undermine credibility.

Mitigation:

- Frontier's guidance: only direct removals count. We measure only the carbon embodied in HAC pellets stored underground, excluding methane avoidance in our credited totals.
- Use mass-balance accounting: weigh feedstock in, pellets out, and track transport emissions precisely.
- Independent third-party verification of data flows.

3.2 Durability Verification

Risk: Reviewers may question >1,000-year permanence of pellets.

Mitigation:

- Provide ultimate analysis demonstrating low H/C and O/C ratios.
- Conduct laboratory stability and leaching tests under simulated mine conditions.
- Use long-term mine monitoring wells for assurance against leachate migration.

3.3 Transparency and Trust

Risk: New CDR methods face skepticism.

Mitigation:

- Publish peer-reviewed data as pilots proceed.
- Engage independent MRV partners (e.g., Puro.earth methodologies).
- Share open data with Frontier and regulators for transparency.

4. Ecosystem and Social Risks

4.1 Community Concerns

Risk: Residents may worry about truck traffic, odors, or storage safety.

Mitigation:

- Roll-off boxes are sealed to prevent odor and spillage.
- Short-haul clustering minimizes traffic impacts.
- Storage involves inert solids, not hazardous liquids or gases.
- Conduct proactive community engagement, town halls, and transparent reporting.

4.2 Nutrient Flows

Risk: Removing manure could deprive farms of soil nutrients.

Mitigation:

- Extract, concentrate, and return nutrients (N, P, K, Ca, Mg) as precision fertilizer.
- Farmers retain or improve agronomic value while decoupling carbon from nutrient cycling.

4.3 Ecosystem Leakage

Risk: If mine voids leaked or collapsed, stored carbon could be released.

Mitigation:

- HAC pellets are insoluble and highly stable; even if disturbed, they remain inert solids.
- HAC pellets mass density is higher than water, and mine sites are below the lowest USDW. Migration upward is highly unlikely.
- Regulatory mine monitoring ensures water quality is tracked for decades.

5. Financial Risks

5.1 Capital Intensity

Risk: Hub deployment requires ~\$5M capex; early-stage capital availability could limit rollout.

Mitigation:

- Use blended financing (equity + USDA REAP loans).
- Stage deployment: 1 t/day demo → 10 t/day pilot → 80 t/day FOAK → replication NOAK.
- Maintain low-risk, modular footprint (containerized TSC units).

5.2 Revenue Dependence on Carbon Credit Markets

Risk: Project economics rely on premium credit prices (\$200-400/tCO₂e).

Mitigation:

- Frontier pre-purchase provides anchor demand.
- Diversify revenue: nutrient products, tipping fees, activated carbon sales where appropriate.
- Declining cost curve with replication ensures competitiveness as markets mature.

5.3 Farmer Compensation Sustainability

Risk: Payments to farmers (\$40/dry tonne) must remain feasible.

Mitigation:

- Modeled into project economics: logistics + farmer payments + opex represent <40% of total cost.
- Early commitments with Frontier reduce revenue uncertainty.

6. Summary

The largest risks for Intercept Carbon lie in project execution: technology scale-up, farmer and mine participation, MRV credibility, and early-stage financing. Technical risks are low, as Thermo-Sonic Carbonization is robust across feedstocks and produces a chemically stable output. Ecosystem and community risks are modest, given the inert nature of HAC pellets and existing mine monitoring frameworks.

Our risk philosophy is clear: design for simplicity, align with farmers, and build transparency into monitoring. By proactively addressing each risk class – technical, execution, MRV, social, and financial – we ensure that our pathway to gigaton-scale carbon removal is durable, credible, and equitable.

- c. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	500t CO ₂ e
Delivery window	Dec 31, 2027

<i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$353
Levelized price (\$/ton CO ₂) ² <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$1,000

² This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).